IRE 203 Introduction to IoT and Robotics

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Internet Origin: History of internet Invention from ARPANET to Modern Web (1957-2025)

The **internet** as we know it today is a vast, interconnected network that has transformed how we communicate, work, learn, and interact. But it wasn't always this way. The origin of the internet dates back to the 1950s when the world was grappling with the Cold War, and researchers sought to create a communication system that could withstand disruptions. From its humble beginnings as a military project to the global web it is now, the **history of the internet** is a tale of human ingenuity, collaboration, and the relentless pursuit of connectivity and a fascinating journey of innovation, collaboration, and rapid technological advancement.

Internet Origin: History of internet Invention from ARPANET to Modern Web (1957-2025)

Key Milestones in Internet History

1969: *Launch of* ***ARPANET****, the first operational packet-switching network.*

1971: *The first* ***email*** *is sent by* ***Ray Tomlinson****.*

1983: *Adoption of* ***TCP/IP*** *as the standard protocol for ARPANET.*

1989-1990: Tim Berners-Lee *invents the* ***World Wide Web****.*

1993: *The first graphical web browser,* ***Mosaic****, is released.*

1998: Google *is founded, revolutionizing search engines.*

2007: The iPhone *introduces widespread mobile Internet access.*

2010s: *Rise of* ***cloud computing****,* ***social media****, and* ***streaming services****.*

Who is known as the father of the Internet?

Vint Cerf and Bob Kahn *Known as "The Fathers of the Internet"*

Key Contributions:

- *Invented TCP/IP protocol in 1974*
- *Created the foundation for data transmission*
- *Developed internet architecture*
- *Established basic internet communication standards*

https://www.geeksforgeeks.org/websites-apps/history-of-internet/

Advanced Research Projects Agency NET

ARPANET stands for **Advanced Research Projects Agency NET**. ARPANET was the first network that consisted of distributed control. It was the first to implement TCP/IP protocols. It was basically the beginning of the Internet with the use of these technologies. It was designed with a basic idea in mind that was to communicate with scientific users among an institute or university.

https://www.techtarget.com/searchnetworking/definition/ARPANET

Advanced Research Projects Agency NET

History of ARPANET

ARPANET was introduced in the year 1969 by the Advanced Research Projects Agency (ARPA) of the US Department of Defense. It was established using a bunch of PCs at various colleges and sharing of information and messages was done. It was for playing as long separation diversions and individuals were asked to share their perspectives. In the year 1980, ARPANET was handed over to a different military network, Defense Data Network.

Advanced Research Projects Agency NET

Characteristics of ARPANET

- It is basically a type of WAN.
- It used the concept of a packet-switching network.
- It used Interface Message Processors(IMPs) for sub-netting.
- ARPANETs software was split into two parts- a host and a subnet.

Advantages of ARPANET

- ARPANET was designed to serve even in a Nuclear Attack.
- It was used for collaborations through E-mails.
- It created an advancement in the transfer of important files and data for defense.

Limitations of ARPANET

- Increased number of LAN connections resulted in difficulty handling.
- It was unable to cope-up with advancement in technology.

Advanced Research Projects Agency NET

Q. Explain how an IP address ensures data reaches the correct device.

Q. Compare centralized and distributed network control.

Q. Describe distributed control in ARPANET with a real-life analogy.

Q. Why did increase the number of LAN connections create difficulties for ARPANET?

WWW, DNS, URL

The **World Wide Web (WWW)** is an information system that allows users to access documents and resources over the internet. A **Uniform Resource Locator (URL)** is the human-readable address used to locate these resources, while the **Domain Name System (DNS)** translates these addresses into numerical IP addresses required for communication between computers.

WWW, DNS, URL

1. WWW (World Wide Web)

i. A system of interconnected web-based documents such as webpages, images, and files.

ii. Accessed through web browsers using **HTTP or HTTPS** protocols.

iii. It is a service that runs on the internet.

2. URL (Uniform Resource Locator)

i. The complete address of a resource on the web.

ii. It tells the browser **how** and **where** to access a resource.

iii. Structure of a URL: https://www.example.com/page.html

https : Protocol

www.example.com : Domain name

/page.html : Path to the resource

WWW, DNS, URL

3. DNS (Domain Name System)

- A hierarchical and distributed system.
- Translates domain names into IP addresses.
- Makes it easier for users to access websites using names instead of numbers.
- **Example:**
- www.example.com → 192.0.2.1

WWW, DNS, URL

How They Work Together?

•Input:

You type a URL in your browser, for example: www.example.com.

The browser needs to find out where this website is on the Internet.

•DNS Lookup:

DNS = Domain Name System → like the Internet's phonebook.

The browser asks DNS: "What is the IP address of www.example.com?"

Resolution:

The DNS server finds the correct IP address for the website.

Example: www.example.com → 192.0.2.1

The browser now knows where to send the request.

Connection:

The browser connects to the web server at that IP address.

This is like dialing the phone number you got from the DNS.

Access via WWW:

The web server sends the requested webpage back to your browser.

You can now see and interact with the website.

WWW, DNS, URL

Q. Describe the working process of a URL from browser input to webpage display.

Q. What was ARPANET and why is it important in the history of the Internet?

Q. Explain the structure of a URL with an example.